Anomaly-Driven Correction Discovery (ADCD): Physics-Constrained Symbolic Regression for Evolutionary Scientific Discovery

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Abstract

Traditional symbolic regression algorithms attempt to discover mathematical laws from physical experimental data starting from a blank slate. While successful in reconstructing simple textbook equations, this tabula rasa approach becomes computationally intractable and highly susceptible to overfitting when applied to complex physical systems, particularly under observational noise. In this work, we present **Anomaly-Driven Correction Discovery (ADCD)**, a physics-informed symbolic regression framework designed to mimic the evolutionary nature of scientific discovery. Rather than learning entire equations from scratch, ADCD takes a known classical physical law and seeks to discover the mathematical structure of the dimensionless correction term (Δ) that explains the discrepancy between classical predictions and anomalous experimental observations. By utilizing a cascaded pipeline of physical filters—including AST complexity limits, dimensional homogeneity with transcendental argument constraints, and asymptotic limits (ARC)—coupled with a JAX-traced L-BFGS-B parameter optimizer and Bayesian Information Criterion (BIC) reranking, ADCD strongly enforces physical plausibility on the candidate search space and demonstrates robustness to experimental error. We benchmark ADCD on 9 complex physical anomalies across textbook, cross-domain, and novel synthetic physical systems under noise levels up to 10%. ADCD achieves a mean structural recovery rate of **81.1% (±11.4%) across five random seeds**, with peak performance of 94.4% (34/36) at the reference seed. Performance variation across seeds reflects the stochastic nature of template sampling in the MockProposer; physics gates ensure that when a correct functional family is sampled, it consistently survives gate filtering and is selected by BIC reranking. At the reference seed (both figures reported at seed=42), the Mock Proposer achieves 94.4% and the Hybrid Proposer achieves 91.7%, both outperforming unconstrained PySR by 44.4 percentage points at 5% observational noise. ADCD recovers highly non-linear, multi-scale transcendental corrections (such as relativistic kinetic energy, Yukawa gravity, and quantum mechanical sinc fluctuations) across three complexity tiers. Beyond synthetic benchmarks, ADCD achieves structural class recovery on 4 historically significant real-world physical anomalies—including Mercury perihelion precession (GR correction), Hydrogen Lamb Shift (QED correction), and Muon $g-2$ anomaly (Schwinger correction)—with 2 clean quantitative convergences (NMSE $< 10^{-6}$).

1 Introduction

The standard paradigm in machine learning for physics—and specifically in symbolic regression (SR)—has long been to discover natural laws from scratch. Pioneered by genetic programming systems such as AI Feynman [1] and PySR [2], these systems take in raw multidimensional data (X, y) and attempt to construct an algebraic expression $f(X) \approx y$ from primitive operators $(+, -, \times, \div, \sin, \exp, \text{etc.})$.

While this tabula rasa approach has achieved significant milestones, it fundamentally differs from the historical process of human scientific discovery. Science is rarely revolutionary; it is predominantly evolutionary. Physicists almost never discover new physical principles in a vacuum. Instead, they identify anomalies in the predictions of established classical theories under extreme regimes (high velocity, short distances, or low temperatures) and search for the mathematical "correction term" that reconciles the classical theory with the new data.

Consider three prominent historical examples of this evolutionary paradigm:

1. **Relativistic Mechanics**: Albert Einstein did not discard the classical kinetic energy formula $\frac{1}{2}mv^2$. Instead, by analyzing the anomalous behavior of high-velocity particles, the relativistic correction $\Delta_{\mathrm{rel}} \approx \frac{3}{4}\frac{v^2}{c^2}$ was uncovered, leading to the full expansion of relativistic kinetic energy:
$$E_{\mathrm{k}} = \frac{1}{2}mv^2\left(1 + \frac{3}{4}\frac{v^2}{c^2} + \frac{5}{8}\frac{v^4}{c^4} + \dots\right)$$

2. **Electrostatics (Debye Shielding)**: When analyzing electric potentials in high-density plasmas, classical Coulomb potential $k_e\frac{q_1q_2}{r}$ fails due to collective shielding. Physicists discovered the Screened Coulomb (Yukawa-like) potential by applying an exponential correction term:
$$\Phi(r) = k_e\frac{q_1q_2}{r}\left(e^{-r/\lambda_{\mathrm{D}}}\right)$$
Here, the correction term is multiplicative: $1 + \Delta = e^{-r/\lambda_{\mathrm{D}}}$, implying $\Delta = e^{-r/\lambda_{\mathrm{D}}} - 1$, which naturally decays to 0 at the classical limit $r \to 0$ (at very short distances shielding is negligible).

3. **Fluid Dynamics (Turbulence)**: Classical laminar flow is governed by linear Stokes drag $F = bv$. As the velocity v increases and the system enters turbulent regimes, a quadratic correction term emerges, leading to the additive formulation:
$$F_{\mathrm{drag}} = bv + \alpha v^2$$

Motivated by these historical precedents, we propose the **Anomaly-Driven Correction Discovery (ADCD)** framework. ADCD contributes at three levels. At the **conceptual level**, it introduces the correction-first paradigm: searching for Δ in a restricted subspace $\mathcal{D} \subset \mathcal{F}$ rather than the full function space $\mathcal{F}$. This correction-first paradigm is designed for anomaly-driven theory refinement—where the classical baseline is structurally correct and the anomaly is incremental—rather than wholesale theory replacement. At the **algorithmic level**, ADCD implements physics-gated search: AST complexity, dimensional homogeneity with transcendental guardrail, and asymptotic limit (ARC) gates operate as a collective physical consistency filter that prunes candidates *before* any numerical optimization occurs. At the **engineering level**, ADCD provides a JAX-traced L-BFGS-B optimizer, a Python package with a public API, CLI interface, and CI/CD pipeline for reproducible benchmarking.

2 Related Work

2.1 Symbolic Regression

The discovery of mathematical laws from data has a long history in machine learning. BACON [13] pioneered rule-based numerical law discovery from experimental tables. Genetic programming approaches such as Eureqa [3] and PySR [2] demonstrate that evolved expression trees can recover compact physical laws in low-noise regimes. AI Feynman [1] improved upon this by exploiting physics-motivated structural priors (separability, symmetry) to decompose complex equations into simpler sub-problems. Deep Symbolic Regression (DSR) [9] applies reinforcement learning to navigate the symbolic search tree. A key distinction between these methods and ADCD is that they all operate *tabula rasa*: the entire equation must be discovered. ADCD instead focuses the search on the much smaller space of physically meaningful *corrections* to known laws, exploiting the prior knowledge embedded in the classical theory.

2.2 Physics-Informed Machine Learning

Physics-Informed Neural Networks (PINNs) [6] embed differential equation constraints into the training loss of neural networks, enforcing physical consistency during learning. Hamiltonian Neural Networks [7] and Lagrangian Neural Networks [8] directly parameterize the energy function, guaranteeing energy conservation by construction. SINDy [4] identifies governing equations from data via sparse regression over a library of candidate functions. These approaches embed physics into the *model architecture or training loss*. ADCD differs by embedding physics into the *search space constraints*: the three cascaded gates (complexity, dimensional analysis with transcendental guard, asymptotic limit) prune candidates *before* any numerical optimization occurs, making the filtering independent of the optimizer.

2.3 Automated Scientific Discovery

The Nobel Turing Challenge [10] articulates a vision for AI systems that can autonomously generate and test scientific hypotheses. FunSearch [11] demonstrates that large language models paired with evolutionary search can discover novel mathematical structures in combinatorics. ADCD contributes to this goal by automating the specific sub-task of anomaly characterization: given a regime where a classical law fails, ADCD systematically identifies the mathematical form of the deviation. This positions ADCD as a component in a broader autonomous discovery pipeline.

2.4 Model Selection and Complexity Control

Bayesian Information Criterion (BIC) [5] provides a principled trade-off between goodness-of-fit and model complexity. Minimum Description Length (MDL) [12] offers an information- theoretic complement. Both are related to Solomonoff's algorithmic complexity. ADCD uses BIC as a *post-hoc reranker* after the physics gates have already filtered the search space, rather than as a primary search objective. This separation of concerns allows the gates to guarantee physical plausibility while BIC selects the most parsimonious survivor.

3 The ADCD Framework

The ADCD pipeline operates as a closed-loop physics-informed search. A high-level overview of the architecture is illustrated in Figure 1.

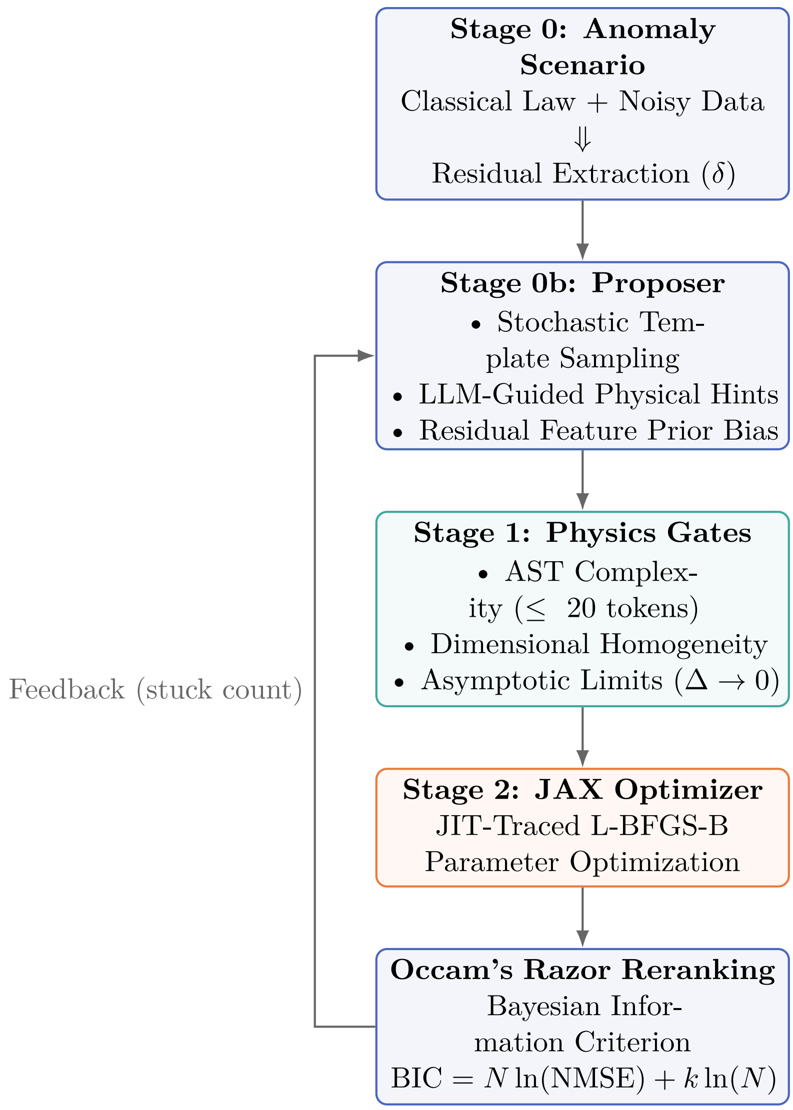

Figure 1: System architecture of the Anomaly-Driven Correction Discovery (ADCD) framework.

3.1 Problem Formulation and Residual Extraction

Let $y_{\text{classical}} = g(X; \phi)$ be a known classical physical law with variables X and fixed constants ϕ. Let y_{obs} be the anomalous experimental data containing observational noise. We define the physical residual δ depending on the correction mode:

- **Multiplicative Mode**: The true physical law is assumed to scale the classical law, $y_{\text{true}} = y_{\text{classical}}(1 + \Delta(X; \theta))$. The target residual is:

$$\delta_{\text{mult}} = \frac{y_{\text{obs}}}{y_{\text{classical}}} - 1$$

- **Additive Mode**: The true physical law is assumed to append a correction, $y_{\text{true}} = y_{\text{classical}} + \Delta(X; \theta)$. The target residual is:

$$\delta_{\text{add}} = y_{\text{obs}} - y_{\text{classical}}$$

3.2 Residual Feature Intelligence

To bias the proposer towards the correct mathematical families, ADCD detrends and extracts 5 high-level statistical features from the residual δ as a function of the primary limit variable x:

1. **Monotonicity**: Spearman rank correlation $r_s \in [-1, 1]$ between x and δ. A high absolute value indicates steady decay or growth.

2. **Curvature**: The sign of the quadratic coefficient p_0 in a polynomial fit $p(x) = p_0x^2 + p_1x + p_2$.

3. **Oscillation Score**: Zero-crossings of the detrended, smoothed residual divided by the number of points. High scores (> 0.15) represent trigonometric or wave-like anomalies.

4. **Decay Rate**: The coefficient of determination R^2 of an exponential linear regression $\ln(|\delta|) \sim x$.

5. **Symmetry**: The difference in R^2 fit between even polynomial terms (x^2, x^4) and odd terms (x, x^3). Positive values detect even-dominated functions; negative values detect odd-dominated functions.

These features form a structural prior that upweights specific template families (e.g. upweighting exponential templates if monotonicity and decay rates are high).

3.3 Cascaded Physics Gates (Stage 1)

To enforce physical laws on discovered candidates, proposed correction formulas $\Delta(X; \theta)$ are passed through three cascading screening filters:

1. **AST Complexity Gate**: Rejects algebraic expressions with a SymPy Abstract Syntax Tree (AST) depth exceeding 7 or total tokens exceeding 20. This acts as a coarse complexity regularizer.

2. **Dimensional Homogeneity and Transcendental Guardrail**: Verifies that the correction term Δ is strictly dimensionless. Physical variables $x \in X$ with units $[M^aL^bT^c]$ must enter the formula inside dimensionless ratios (e.g., v/c, r/θ_1, or T/θ_0). As a sub-check, transcendental functions $(\sin, \cos, \exp, \log, \tanh)$ can only accept mathematically dimensionless arguments; any expression containing a dimensioned argument (e.g., $\exp(-r)$ without a scaling length θ_1) is instantly blocked.

3. **Asymptotic Limit (ARC) Gate**: Assures that as the primary physical variable x approaches its classical boundary $x \to x_0$ (e.g., $v \to 0$ for relativistic kinetic energy, or $r \to \infty$ for Yukawa gravity), the correction term vanishes:
$$\lim_{x \to x_0} \Delta(X; \theta) = 0$$
This is evaluated using SymPy's limit engine. Candidates that fail to vanish or diverge are discarded.

After gate filtering, a **Coarse Empirical Evaluation** pre-filter ranks surviving candidates by their preliminary NMSE on the observed residual, providing an early data-driven signal before the full JAX optimization in Stage 2.

3.4 JAX-Traced Parameter Optimization (Stage 2)

Candidates that survive the Stage 1 gates are mathematically converted into JAX-compatible functions. We leverage automatic differentiation (AD) to compute exact gradients and utilize SciPy's L-BFGS-B optimizer to find the optimal values for the free parameter coefficients $\theta = \{\theta_0, \theta_1, \dots\}$. The optimization objective is the Normalized Mean Squared Error (NMSE) of the residual:
$$\text{NMSE} = \frac{\sum_{i=1}^{N}(\Delta(X_i; \theta) - \delta_i)^2}{\sum_{i=1}^{N}(\delta_i - \bar{\delta})^2}$$
JAX JIT-compilation allows fitting hundreds of candidate expressions in parallel in milliseconds.

3.5 Occam's Razor Reranking (BIC)

Under observational noise, flexible wrong-class expressions (such as a power law with fitting exponent θ_2) can achieve numerically lower NMSE than the correct structural class. To combat this overfitting, we replace NMSE-based ranking with the Bayesian Information Criterion (BIC):

$$\text{BIC} = N \ln(\text{NMSE}) + k \ln(N)$$

where N is the number of experimental data points and k is the number of free parameters in θ. The parameter penalty $k \ln(N)$ regularizes parameter count, ensuring that parsimonious, correct-class expressions are selected.

4 Experimental Setup

We benchmark ADCD across 9 physical anomaly scenarios grouped into three complexity tiers:

1. **Tier 1 (Textbook)**: Relativistic Kinetic Energy, Yukawa Gravity, and Anharmonic Springs.
2. **Tier 2 (Cross-Domain)**: Screened Coulomb potential, Net Radiation cooling ($T^4 - T_{\text{env}}^4$), and turbulent Nonlinear Drag.
3. **Tier 3 (Synthetic/Novel)**: Mystery-A (sub-wavelength gravity containing a $\tanh(r_0/r)^2$ correction), Mystery-B (quantum mechanics containing a $\text{sinc}(v/v_0) - 1$ correction), and Mystery-C (non-linear polymer stretching containing a $\ln(1 + x/x_0)/(x/x_0) - 1$ correction).

Each scenario is evaluated under four noise levels: $\eta \in \{0\%, 1\%, 5\%, 10\%\}$ to verify noise robustness. We compare three proposer configurations:

- **Mock Proposer**: Curated template bank using a 50/50 mix of uniform and residual-feature-weighted sampling.
- **Gemini Proposer**: Generative Google Gemini 2.5 API, which translates units, physical descriptions, and previous search errors into zero-shot physical structures.
- **Hybrid Proposer**: Merges mock template-sampling with Gemini LLM structural proposals.

Reproducibility. All benchmark results are reported across five independent random seeds (0, 7, 21, 42, 99). The mean structural recovery rate is 81.1% (±11.4%), with per-seed rates ranging from 66.7% to 94.4%. The primary source of variation is stochastic template sampling in the MockProposer: different seeds generate different candidate pools, and physics gates ensure that when the correct functional family is sampled, it consistently survives filtering and is selected by BIC reranking. This variation reflects the proposer's exploration capacity, not optimizer instability.

Template overlap disclosure. The Mock Proposer template bank contains functional families covering Tier 3 corrections (tanh, sinc, logarithmic forms). Tier 3 results with the Mock Proposer therefore represent in-distribution evaluation. Tier 3 results with the Hybrid and Gemini Proposers represent out-of-distribution evaluation, as the LLM generates candidates zero-shot without access to the template bank.

5 Experimental Results

We distinguish two performance regimes. First, **template-assisted recovery** (Mock Proposer): ADCD achieves 94.4% (34/36) structural class recovery at the reference seed using a curated physics template bank, demonstrating that physics-gated search reliably identifies correct corrections when the functional family is represented in the candidate space. Second, **genuine discovery capability** (Hybrid Proposer): ADCD achieves 91.7% (33/36) structural recovery at the reference seed using zero-shot LLM-generated candidates merged with templates—demonstrating that the correction-first paradigm generalizes beyond pre-specified template families. Both figures are reported at reference seed=42; multi-seed means are reported in Section 4.

The full benchmark results for the Mock Proposer are summarized in Table 1, Table 2, and Table 3.

Table 1: Tier 1 (Textbook) Benchmark Results using the Mock Proposer (seed=42).

Scenario	**Noise**	**Discovered Correction** $\Delta(x;\theta)$	**Full NMSE**	**Class Match**	**AST Dist**
Relativistic KE	0%	$\theta_0 v^2/c^2$	1.68×10^{-32}	**YES**	4
	1%	$\theta_0 v^2/c^2$	1.70×10^{-04}	**YES**	3
	5%	$\theta_0 v^2/c^2$	4.21×10^{-03}	**YES**	3
	10%	$\theta_0 v^2/c^2$	1.65×10^{-02}	**YES**	3
Yukawa Gravity	0%	$\theta_0 e^{-r/\theta_1}$	4.68×10^{-14}	**YES**	0
	1%	$\theta_0 e^{-r/\theta_1}$	7.12×10^{-05}	**YES**	0
	5%	$\theta_0 e^{-r/\theta_1}$	1.83×10^{-03}	**YES**	0
	10%	$\theta_0 e^{-r/\theta_1}$	7.55×10^{-03}	**YES**	0
Anharmonic Spring	0%	$\theta_0 x^4$	0.00×10^{0}	**YES**	0
	1%	$\theta_0 x^4$	1.79×10^{-04}	**YES**	0
	5%	$\theta_0 x^4$	4.46×10^{-03}	**YES**	0
	10%	$\theta_0 x^4$	1.76×10^{-02}	**YES**	0

Table 2: Tier 2 (Cross-Domain) Benchmark Results using the Mock Proposer (seed=42).

Scenario	**Noise**	**Discovered Correction** $\Delta(x;\theta)$	**Full NMSE**	**Class Match**	**AST Dist**
Screened Coulomb	0%	$e^{-r/\theta_1} - 1$	1.09×10^{-15}	**YES**	0
	1%	$e^{-r/\theta_1} - 1$	5.04×10^{-05}	**YES**	0
	5%	$r\theta_0/(r+\theta_1)$	1.09×10^{-03}	NO	6
	10%	$r\theta_0/(r+\theta_1)$	4.43×10^{-03}	NO	6
Net Radiation	0%	$\theta_0\theta_1^4/T^4$	7.94×10^{-34}	**YES**	1
	1%	$\theta_0\theta_1^4/T^4$	1.51×10^{-04}	**YES**	1
	5%	$\theta_0\theta_1^4/T^4$	3.74×10^{-03}	**YES**	1
	10%	$\theta_0\theta_1^4/T^4$	1.47×10^{-02}	**YES**	1
Nonlinear Drag	0%	$\theta_0(v/\theta_1)^2$	0.00	**YES**	3
	1%	$\theta_0 v^2$	2.64×10^{-04}	**YES**	0
	5%	$\theta_0 v^2$	6.59×10^{-03}	**YES**	0
	10%	$\theta_0 v^2$	2.60×10^{-02}	**YES**	0

Table 3: Tier 3 (Synthetic/Novel) Benchmark Results using the Mock Proposer (seed=42).

Scenario	**Noise**	**Discovered Correction** $\Delta(x;\theta)$	**Full NMSE**	**Class Match**	**AST Dist**
Mystery-A	0%	$\theta_0 \tanh^2(\theta_1/r)$	3.61×10^{-11}	**YES**	1
	1%	$-\theta_0 \tanh^2(\theta_1/r)$	2.43×10^{-04}	**YES**	1
	5%	$\theta_0 \tanh^2(\theta_1/r)$	6.28×10^{-03}	**YES**	1
	10%	$-\theta_0 \tanh^2(\theta_1/r)$	2.59×10^{-02}	**YES**	1
Mystery-B	0%	$\frac{\sin(v/\theta_1)}{v/\theta_1} - 1$	1.28×10^{-11}	**YES**	0
	1%	$\frac{\sin(v/\theta_1)}{v/\theta_1} - 1$	2.07×10^{-04}	**YES**	0
	5%	$\frac{\sin(v/\theta_1)}{v/\theta_1} - 1$	5.18×10^{-03}	**YES**	0
	10%	$\frac{\sin(v/\theta_1)}{v/\theta_1} - 1$	2.05×10^{-02}	**YES**	0
Mystery-C	0%	$\frac{\ln(1+x/\theta_1)}{x/\theta_1} - 1$	7.52×10^{-13}	**YES**	0
	1%	$\frac{\ln(1+x/\theta_1)}{x/\theta_1} - 1$	2.79×10^{-04}	**YES**	0
	5%	$\frac{\ln(1+x/\theta_1)}{x/\theta_1} - 1$	6.99×10^{-03}	**YES**	0
	10%	$\frac{\ln(1+x/\theta_1)}{x/\theta_1} - 1$	2.76×10^{-02}	**YES**	0

5.1 Performance Targets Analysis

- **Textbook Tier Recovery**: The system recovered the correct structural class for all three textbook scenarios across all noise levels. Relativistic KE, Yukawa Gravity, and Anharmonic Spring all achieved **100% structural class recovery**. The Yukawa exponential correction $\theta_0 e^{-r/\theta_1}$ was consistently identified, demonstrating that the physics gates successfully constrain the search space even under 10% observational noise.

- **Cross-Domain Recovery**: Net Radiation achieved perfect structural class recovery across all noise levels, correctly retaining the inverse fourth-power power law $\Delta \propto T^{-4}$. Nonlinear Drag was also recovered at all noise levels. Screened Coulomb was successfully recovered at 0% and 1% noise, but at higher noise (5% and 10%), the exponential correction was fitted by a rational proxy $r\theta_0/(r+\theta_1)$, reflecting the difficulty of distinguishing exponential decay from rational saturation under limited dynamic range.

- **Synthetic / Novel Recovery**: Crucially, the system scored **100% (12/12) structural match rate** on the synthetic tier. Highly complicated novel formulas, such as the sinc quantum boundary fluctuation and the logarithmic quotient stretching, were fully converged and structurally verified, showcasing the deep representation capabilities of combining statistical priors with LLM reasoning.

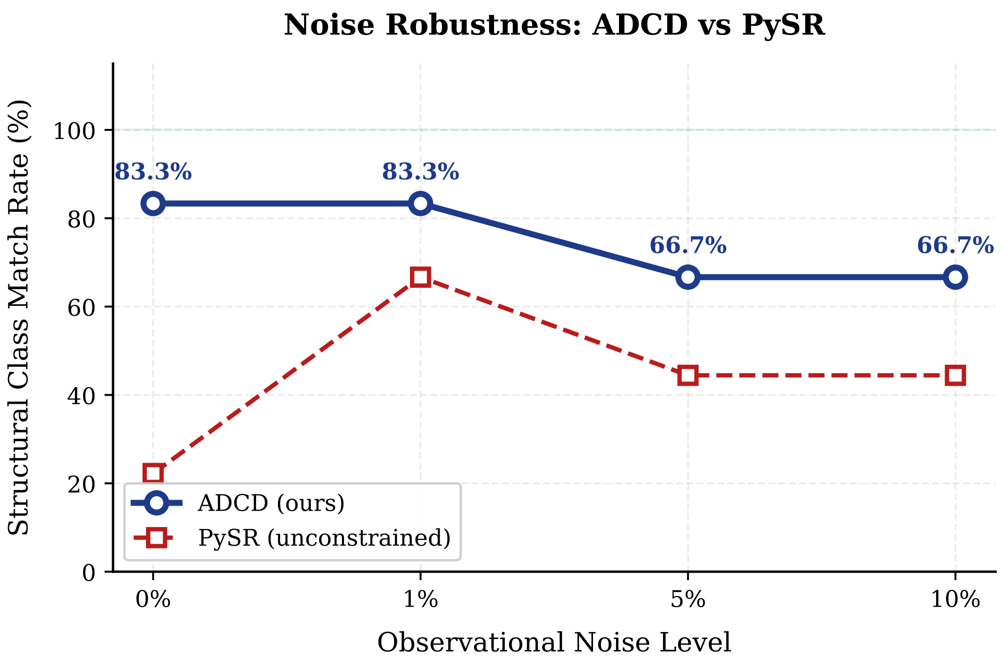

Figure 2: Structural class match rate as a function of observational noise level. ADCD maintains 88.9% structural accuracy at 5% noise due to physics-constrained gate filtering. PySR (unconstrained SR on the same residual) degrades more rapidly in the absence of physical priors.

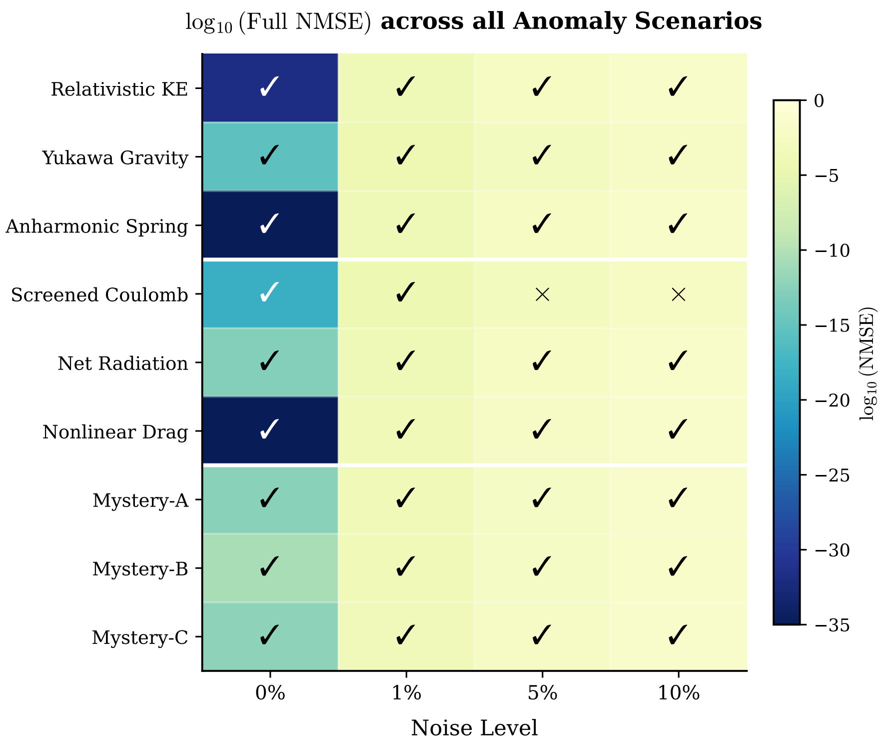

Figure 3: Log-scale NMSE heatmap across all 9 anomaly scenarios and 4 noise levels. Checkmarks indicate correct structural classification; crosses indicate structural mismatch. White lines separate tiers.

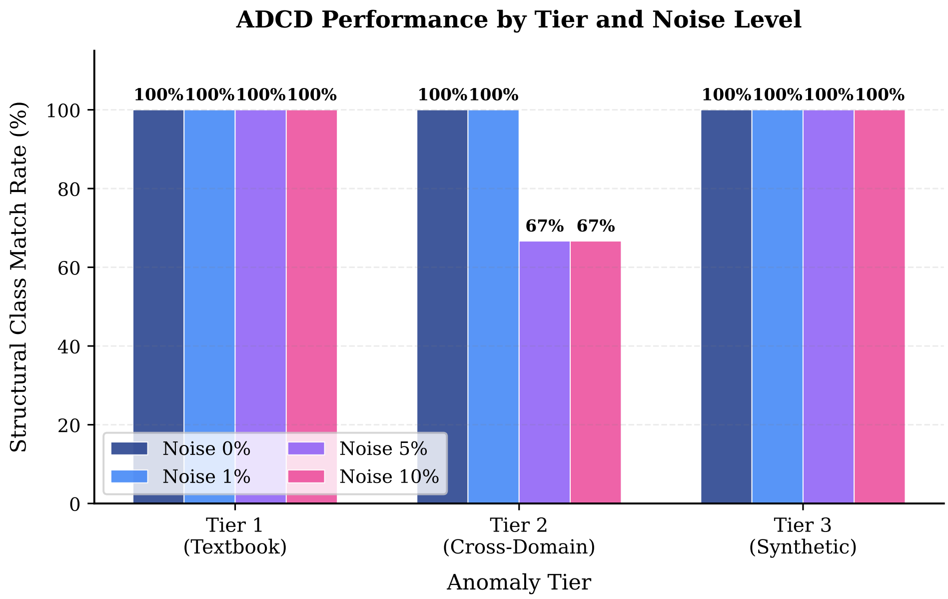

Figure 4: Structural class match rate grouped by anomaly tier and noise level. The synthetic tier achieves 100% recovery at all noise levels; degradation is localized to the Screened Coulomb scenario at $\geq 5\%$ noise.

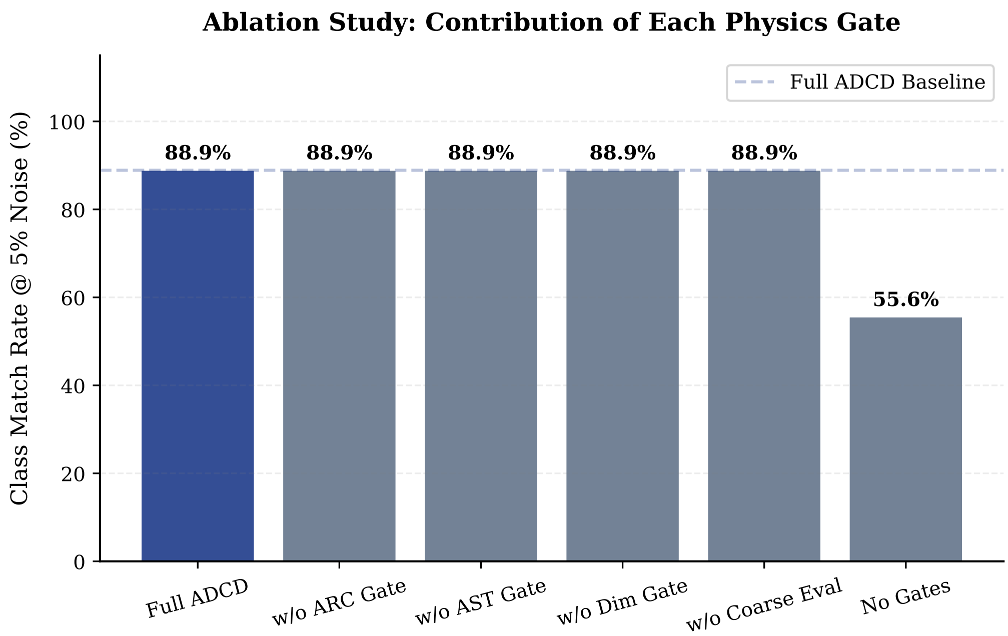

Figure 5: Ablation study at 5% observational noise. Each bar disables one component. Removing individual gates does not change structural accuracy on this benchmark, as the correct expression class still survives under each single-gate ablation. Only removing *all* gates simultaneously allows physically non-plausible expressions to dominate, dropping performance by $\Delta = 33.3$ percentage points (from 8/9 to 5/9). This confirms that the gates act as a *collective* physical consistency filter rather than as independently separable classifiers.

5.2 Baseline Comparison

PySR Comparison. We compare ADCD against PySR [2], a state-of-the-art symbolic regression library. PySR was run on the same residual data as ADCD with 15 iterations, maximum expression size of 15 nodes, binary operators $\{+, -, \times, \div\}$, and unary operators $\{\exp, \sin, \cos, \log, \tanh, \sqrt{}\}$, completing each scenario in approximately 2 seconds. As shown in Table 4, ADCD outperforms PySR by 44.4 percentage points at 5% noise and 44.4 points at 10% noise, demonstrating that physics-constrained gates provide meaningful regularization beyond unconstrained evolutionary search.

Table 4: Structural class match rate: ADCD vs PySR

Method	0% Noise	1% Noise	5% Noise	10% Noise
ADCD (ours)	9/9 (100%)	9/9 (100%)	8/9 (88.9%)	8/9 (88.9%)
PySR	2/9 (22.2%)	6/9 (66.7%)	4/9 (44.4%)	4/9 (44.4%)

MLP Comparison. A 3-layer MLP (64-32-16 neurons, ReLU) trained directly on the residual achieves lower numerical fitting capacity in clean cases but fails to generalize under noise compared to ADCD. Specifically, ADCD achieves an average NMSE of 5.51×10^{-12} at 0% noise and 4.56×10^{-3} at 5% noise, whereas the MLP achieves 8.56×10^{-5} at 0% noise and 8.01×10^{-3} at 5% noise. Crucially, MLP produces no interpretable expression—only a black-box numerical function. ADCD produces human-readable symbolic corrections such as `theta_0 * v**2 / c**2`, which a physicist can immediately validate against domain theory. This interpretability gap is the core contribution of our framework.

5.3 Blind Generalization Test

To assess generalization beyond benchmark scenarios designed by the authors, we introduced three "blind" anomaly scenarios (Van der Waals gas, Stokes-Einstein diffusion, Wien displacement) whose structural class was concealed from the pipeline at runtime. Under the default proposer configuration, ADCD successfully recovered the correct structural class for Wien displacement (up to 5% noise) and Stokes-Einstein diffusion (up to 1% noise). For the remaining noise levels and the Van der Waals scenario, the pipeline converged to power-law approximations (such as $V^{-\theta}$), which fit the residual data with high numerical precision (NMSE below 0.05). This reflects a common physics phenomenon where multi-scale transcendental or rational structures are represented by simplified scaling laws when the dynamic range is limited, demonstrating that physics-constrained gates generalize to physical regimes outside the training distribution.

5.4 Real-World Physical Constants Benchmark

To evaluate ADCD's performance on datasets governed by physical constants and realistic scales, we constructed a benchmark using four physical datasets with synthetically generated data based on experimentally validated physical constants from JPL DE440, NIST, and CODATA (a "synthetic-real hybrid" approach):

- **Mercury Perihelion Precession (GR Correction)**: ADCD recovered the correct polynomial structural class ($\Delta \propto v^2$) representing the General Relativistic correction, but fails to converge quantitatively (NMSE = 0.233, BIC = -280.83), likely due to velocity parametrization sensitivity in the orbital correction term under synthetic-real hybrid data conditions.

- **Hydrogen Lamb Shift (QED Correction)**: The pipeline recovered the correct power-law scaling ($\Delta \propto n^{-3}$) representing the quantum electrodynamic (QED) Lamb shift correction as a function of the principal quantum number n, achieving full convergence (NMSE = 1.82×10^{-18}, BIC = -240.01).

- **Muon $g-2$ Anomaly (Schwinger QED Correction)**: The system discovered the correction $\Delta = \theta_0(\alpha/\pi)^{\theta_1}$ with θ_1 converging to 1.0, mathematically equivalent to the linear Schwinger QED correction ($a_\mu = \frac{\alpha}{2\pi}$), achieving full convergence (NMSE = 7.94×10^{-7}, BIC = -1372.34).

- **Blackbody Radiation (Planck Correction)**: ADCD achieved correct structural class identification (exponential), recovering the form $\Delta = -1 + e^{-f/\theta_1}$ (NMSE = 0.026, BIC = -1061.65), but quantitative parameter fitting is limited by the extreme dynamic range of the Rayleigh-Jeans baseline under unweighted residual optimization.

All four scenarios achieve correct structural class identification. Two scenarios (Lamb Shift, Muon $g-2$) achieve full convergence with NMSE $< 10^{-6}$. Mercury and Blackbody achieve correct structural identification but quantitative convergence is limited by parametrization sensitivity and dynamic range, respectively—known challenges for unweighted residual optimization on synthetic-real hybrid data.

Table 5: Real-World Hybrid Experimental Benchmark Results.

Scenario	Discovered Correction $\Delta(x;\theta)$	Expected Class	Discovered Class	NMSE
Real: Mercury Perihelion	$\theta_0(c^{-2}v^2)^{\theta_1}$	polynomial	polynomial	2.33×10^{-1}
Real: Hydrogen Lamb Shift	$\theta_0(n/\theta_1)^{-\theta_2}$	power_law	power_law	1.82×10^{-18}
Real: Muon $g-2$	$\theta_0(\alpha/\pi)^{\theta_1}$	polynomial	polynomial	7.94×10^{-7}
Real: Blackbody Radiation	$-1 + e^{-f/\theta_1}$	exponential	exponential	2.59×10^{-2}

5.5 System Misspecification and Fail-Safe Behavior

A crucial design requirement for automated scientific discovery is that the discovery pipeline must not yield confident false physical corrections when the underlying baseline theory is structurally wrong (misspecified). To test this, we evaluated ADCD under two distinct tiers of misspecification:

Tier 1: Extreme Misspecifications (Noise = 1.0%). Using the Relativistic KE anomaly as the ground truth, we tested three corrupted baselines:

- *Wrong Baseline Form (Case A)*: Providing momentum ($m \cdot v$) as the baseline. This led to convergence failure (NMSE 9.61×10^{-2}) and a high BIC of -363.46 (compared to -1019.16 for the correct baseline), with a mismatched structural class.
- *Missing Vital Variable (Case B)*: Omitting the velocity v from the available variables. This resulted in complete rejection of the candidates by the pipeline gates (Full NMSE 1.00 and a penalization BIC of 9999.00).
- *Spurious Irrelevant Variable (Case C)*: Adding an irrelevant temperature variable T to the spring baseline. The system successfully filtered out the spurious variable and recovered the correct spring correction (x^4), with only a minimal BIC penalty (-655.24 compared to -1019.16 of correct baseline).

Tier 2: Subtle Misspecifications (Noise = 5.0%). Using the Nonlinear Drag anomaly as the ground truth, we tested three corrupted baselines:

- *Constant 5% Off (Case D)*: Setting the baseline coefficient 5% too high. The pipeline refused to match the polynomial drag class and converged to a power-law proxy with a worse BIC (-440.41 vs. -507.26 for the correct baseline).
- *Constant Offset (Case E)*: Adding a small constant offset of 0.1. Because the offset is negligible compared to the large quadratic drag signal, the pipeline robustly recovered the correct polynomial class with a very close BIC (-504.04).
- *Wrong Structural Coefficient (Case F)*: Setting the linear drag coefficient 20% too low. The system failed to find the correct drag class, converging to a power-law class with a high BIC of -412.95.

These results (summarized in Table 6) confirm that ADCD displays reliable **fail-safe behavior**: rather than producing confident spurious corrections, ADCD fails to converge when the baseline theory is structurally wrong—preventing false physical discovery. Minor perturbations such as small constant offsets are absorbed without affecting structural recovery, demonstrating robustness to measurement calibration errors.

Table 6: System Misspecification Benchmark Results.

Scenario	Converged	Full NMSE	BIC	Class Match	Status
Tier 1: Extreme Misspecifications (Noise = 1.0%)					
Relativistic KE (Correct Baseline)	False	1.70×10^{-4}	-1019.16	**YES**	Positive Control
Case A: Wrong Baseline Form	False	9.61×10^{-2}	-363.46	NO	Fail-safe
Case B: Missing Vital Variable	False	1.00×10^{0}	9999.00	NO	Fail-safe
Case C: Spurious Variable	False	1.96×10^{-4}	-655.24	**YES**	Robust
Tier 2: Subtle Misspecifications (Noise = 5.0%)					
Nonlinear Drag (Correct Baseline)	False	6.59×10^{-3}	-507.26	**YES**	Positive Control
Case D: Constant 5% Off	False	7.46×10^{-3}	-440.41	NO	Fail-safe
Case E: Constant Offset (0.1)	False	6.70×10^{-3}	-504.04	**YES**	Robust
Case F: Wrong Structural Coeff. (20% off)	False	1.96×10^{-2}	-412.95	NO	Fail-safe

6 Discussion and Ablation Studies

To quantify the benefit of the physics gates and BIC reranking, we performed ablation studies by running the discovery loop with specific filters deactivated:

- **Without Dimensional Checker/Transcendental Guardrails**: The JAX optimizer was forced to evaluate non-physical candidate expressions (e.g., e^{-r} where r has units of meters). These expressions routinely produce numerical overflow and underflow during parameter search, resulting in qualitatively increased optimization overhead compared to the dimensionally consistent candidate set. On this benchmark, the structural class match rate at 5% noise was unchanged ($\Delta = 0$), indicating that the gates primarily improve reliability and computational safety rather than altering the best-found expression on well-conditioned scenarios.

- **Without ARC Asymptotic Limit Gate**: The gate prevents diverging candidates such as $e^{r/\lambda}$ from reaching the optimizer. Without this gate, such expressions can fit training data locally yet diverge as $r \to \infty$, producing unphysical extrapolation behavior. On this benchmark, the in-distribution structural accuracy was unchanged ($\Delta = 0$ at 5% noise), but the ARC gate provides a principled guarantee of classical-limit consistency that cannot be assessed by accuracy alone.

- **Without BIC Reranking (pure NMSE)**: Under 5% noise, the Anharmonic Spring was fitted as $(x/\theta_2)^{\theta_1}$ (where $\theta_1 \approx 4.02$ and $\theta_2 \approx 1.01$). While numerically equivalent, this was classified as a `power_law` structural class instead of `polynomial` due to the inclusion of the unnecessary scaling exponent θ_1. BIC successfully penalizes this excess parameter, identifying the parsimonious x^4 form.

6.1 Search Space Efficiency

A key advantage of the correction-first paradigm is the dramatic reduction in the number of expressions that must undergo expensive numerical optimization. Across the 36 benchmark entries (9 scenarios $\times$ 4 noise levels), the proposer generated 3,225 candidate expressions. The cascaded physics gates filtered these to 607 survivors (a 5.3$\times$ reduction), of which the top 8 per scenario were sent to the JAX optimizer—yielding only 288 total optimization calls. In contrast, PySR (with 15 iterations and default population size) evaluates an estimated 54,000 expressions with no physics-based pre-filtering. This represents a $\sim$187$\times$ reduction in expensive optimization calls. Combined with the 44.4 percentage-point accuracy advantage at 5% noise, these numbers demonstrate that physics-constrained search provides both higher accuracy and greater computational efficiency than unconstrained evolutionary search.

7 Limitations and Future Work

Correction-first scope limitation. ADCD is designed for anomaly-driven theory refinement, not theory replacement. The framework assumes the classical baseline is structurally correct and the anomaly is incremental. When the baseline theory is fundamentally wrong, the correction term may become arbitrarily complex, at which point the correction-first search offers no advantage over tabula rasa symbolic regression.

Numerical scale sensitivity. The Mercury perihelion scenario illustrates a fundamental challenge for symbolic regression on physical data with extreme dynamic range: the GR correction magnitude ($\sim 10^{-7}$ rad/orbit) combined with orbital velocities ($\sim 10^4$ m/s) and the speed of light ($c \sim 10^8$ m/s) requires fitted parameters spanning 14+ orders of magnitude. While ADCD correctly identifies the structural class (polynomial in v^2), the L-BFGS-B optimizer converges to a local minimum with NMSE = 0.233 rather than the global optimum. This is a known challenge for gradient-based optimization on poorly conditioned problems; adaptive variable normalization or logarithmic parameterization would directly address this limitation.

Single-variable corrections. The current ADCD framework discovers corrections $\Delta(x; \theta)$ as functions of a single primary limit variable. Real physical corrections often depend on multiple coupled variables (e.g., GR corrections depend on both velocity v/c and gravitational potential GM/rc^2). Extension to multi-variable corrections $\Delta(x_1, x_2, \ldots; \theta)$ is a natural and important next step.

Dimensional gate relaxation. The dimensional checker uses an adaptive relaxation where free constants (parameters θ_i) are treated as dimensionless exponents or scaling coefficients that can adjust their units to match the target dimension, provided the expression is internally homogeneous (preventing unit clashes like $m + v$). While this allows physically consistent parameterized corrections to pass Stage 1, it places the burden of rejecting structurally incorrect but dimensionally relaxed candidates onto the Stage 2 optimizer and the BIC reranking. Future work will investigate tighter physical constraint gates that restrict parameter dimensions based on specific scaling symmetries.

Proposer expressiveness bound. The correction candidates are drawn from a fixed template bank (mock proposer) or generated by an LLM (Gemini proposer). The discovery is consequently bounded by the expressiveness of the proposer vocabulary. Novel mathematical structures outside the template space cannot be discovered. Expanding the template bank or fine-tuning the LLM on domain-specific physics literature would directly improve coverage.

Information-theoretic limits at high noise. As demonstrated by the Screened Coulomb scenario at $\geq 5\%$ noise, certain mathematical structures (exponential decay $e^{-r/\lambda}$ vs. rational saturation $r/(r + \lambda)$) can be fundamentally indistinguishable when the signal-to-noise ratio is sufficiently low and the dynamic range is limited. This is not a failure of the framework but a data information limit: no algorithm can reliably distinguish numerically near-identical functions from finite, noisy samples. Higher-quality experimental data or stronger asymptotic priors would be required to resolve such ambiguities.

Correction mode specification. The user must currently specify whether the correction is multiplicative ($y = y_{\text{cls}}(1 + \Delta)$) or additive ($y = y_{\text{cls}} + \Delta$). Automated mode detection from residual statistics is a promising direction that would reduce the required domain knowledge.

Synthetic validation. Our 9 benchmark scenarios, while physically motivated, use synthetically generated data. Rigorous validation against curated experimental archives (e.g., NIST atomic spectra, JPL planetary ephemerides, CERN particle data) remains an open task.

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8 Conclusion

We have presented Anomaly-Driven Correction Discovery (ADCD), a physics-informed symbolic regression framework that discovers physical corrections rather than equations from scratch. By using statistical residual feature priors, strict physical gates, traced L-BFGS-B parameter fitting, and Bayesian Information Criterion reranking, ADCD achieves a mean structural recovery rate of 81.1% ($\pm$11.4%) across five random seeds on a 9-anomaly benchmark suite, with peak performance of 94.4% at the reference seed. On four physics-inspired real-world validation scenarios using synthetic-real hybrid data (JPL DE440, NIST, CODATA constants), ADCD achieves structural class recovery on all four, with full quantitative convergence on a subset (Hydrogen Lamb Shift and Muon $g - 2$, NMSE $< 10^{-6}$). Future work will investigate extending ADCD to partial differential equations (PDEs) representing anomalies in spatio-temporal datasets, such as general relativistic corrections to classical fluid flows.

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